



百种机器学习联合虚拟细胞基因敲除

(网络毒理学+机器学习+单细胞基因敲除+分子对接)

--生信自学网



主要内容

- 简介
- GEO数据下载和整理
- 差异分析
- 单细胞数据下载和整理
- 细胞聚类和细胞注释
- 化合物靶基因(Diethyl phthalate, DEP)
- 交集基因
- 构建化合物->基因->疾病调控网络
- 构建机器学习诊断模型(113种)
- 选择最优的模型(AUC值)
- 模拟基因敲除分析
- GO和KEGG富集分析
- 分子对接分析

软件准备



GEO数据下载

 NCBI Resources  How To 

[GEO Home](#) | [Documentation](#)  | [Query & Browse](#)  | [Email GEO](#)

[Sign in to NCBI](#)

Gene Expression Omnibus

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.



Getting Started

[Overview](#)[FAQ](#)[About GEO DataSets](#)[About GEO Profiles](#)[About GEO2R Analysis](#)[How to Construct a Query](#)[How to Download Data](#)

Tools

[Search for Studies at GEO DataSets](#)[Search for Gene Expression at GEO Profiles](#)[Search GEO Documentation](#)[Analyze a Study with GEO2R](#)[Studies with Genome Data Viewer Tracks](#)[Programmatic Access](#)[FTP Site](#)

Browse Content

[Repository Browser](#)[DataSets:](#) 4348[Series:](#)  118925[Platforms:](#) 20174[Samples:](#) 3251445

<https://www.ncbi.nlm.nih.gov/geo/>

搜索关键字: Alzheimer disease

GEO数据注释



geneNames	GSM119615	GSM119616
1007_s_at	16.09267	34.99131
1053_at	13.56297	40.78045
117_at	315.3579	133.1235
121_at	45.06876	23.77109
1255_g_at	41.56298	30.73738
1294_at	17.86836	22.18804
1316_at	5.341058	35.84349
1320_at	47.33934	16.27059
1405_i_at	11.98004	52.23917
1431_at	1002.45	611.6513



geneNames	GSM119615	GSM119616
A1BG	16.09267	34.99131
A1CF	13.56297	40.78045
A2M	315.3579	133.1235
A2ML1	45.06876	23.77109
A2MP1	41.56298	30.73738
A4GALT	17.86836	22.18804
A4GNT	5.341058	35.84349
AA06	47.33934	16.27059
AAAS	11.98004	52.23917
AACS	1002.45	611.6513

单个数据矫正

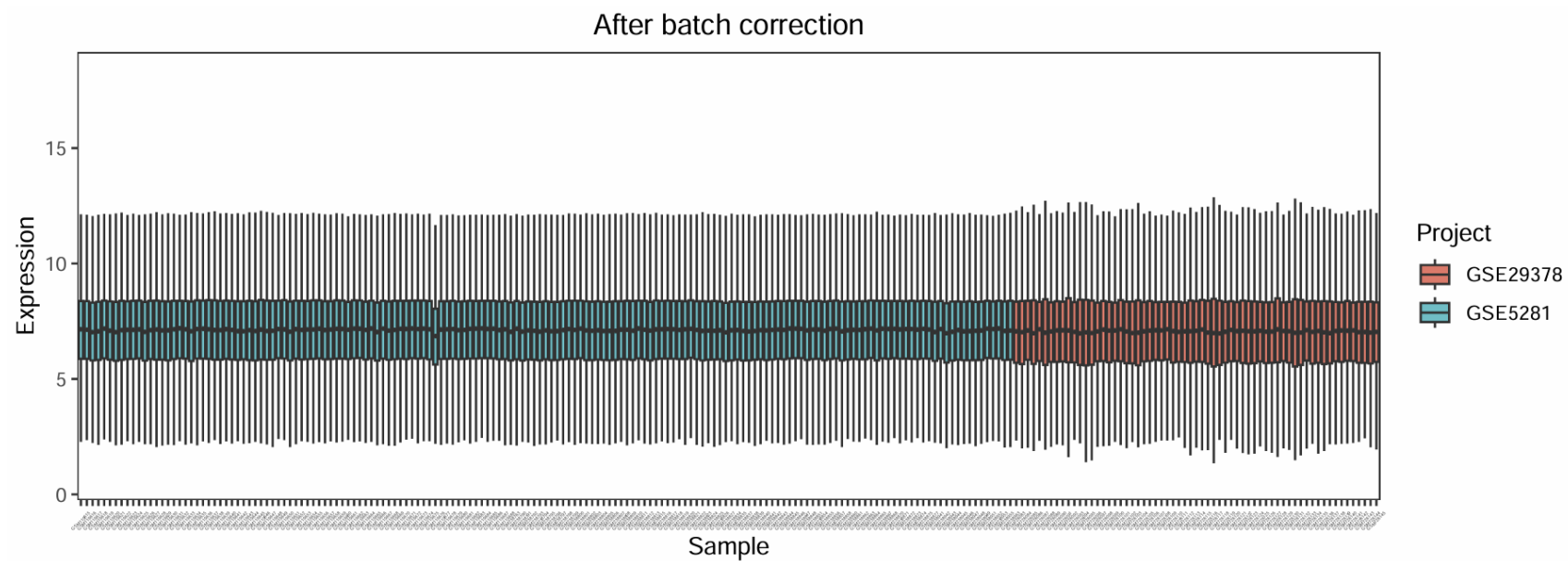
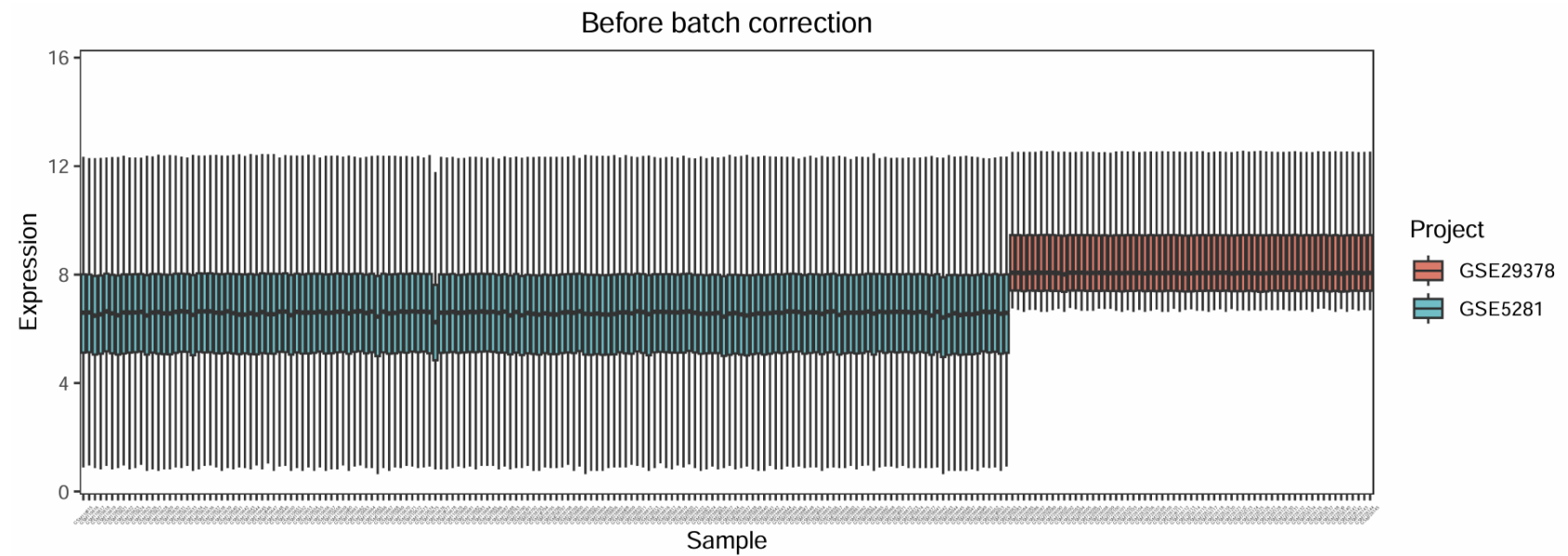
id	GSM119615_ Control	GSM119616_ Control	GSM238763_ Treat	GSM238790_ Treat
A1BG	4.232868	5.320441	2.974238	2.029496
A1BG-AS1	2.562557	6.197063	4.146929	5.859998
A1CF	3.961824	5.518505	7.129386	5.681319
A2M	8.315195	7.108894	9.903188	6.976466
A2M-AS1	6.939425	6.420417	7.332132	6.319804
A2ML1	5.757693	4.78636	6.614038	2.925607
A2MP1	5.663091	5.15545	4.195269	6.022404
A4GALT	4.408479	4.687182	7.34165	3.469129
A4GNT	2.616021	5.351876	7.284282	3.162274
AA06	5.816476	4.23062	6.521356	6.131143

数据合并



geneNames	GSE29378_GSM72 6117_Control	GSE29378_GSM72 6126_Treat	GSE5281_GSM11 9615_Control	GSE5281_GSM23 8763_Treat
A1BG	6.015995	6.694689	4.991165	3.904787
A1CF	6.162944	5.021466	4.712097	7.445556
A2M	8.734031	9.127091	9.128365	10.51673
A2ML1	4.919883	4.720141	6.39502	7.136111
A4GALT	4.341111	6.998158	5.354791	7.898451
A4GNT	4.813435	5.772101	3.674817	7.707595
AAAS	6.87463	4.900647	4.723635	4.709897
AACS	10.58925	9.901191	9.729013	9.638343
AADAC	4.40292	3.955708	4.831234	3.363282
AADACL2	5.809605	5.4465	5.875845	4.980885

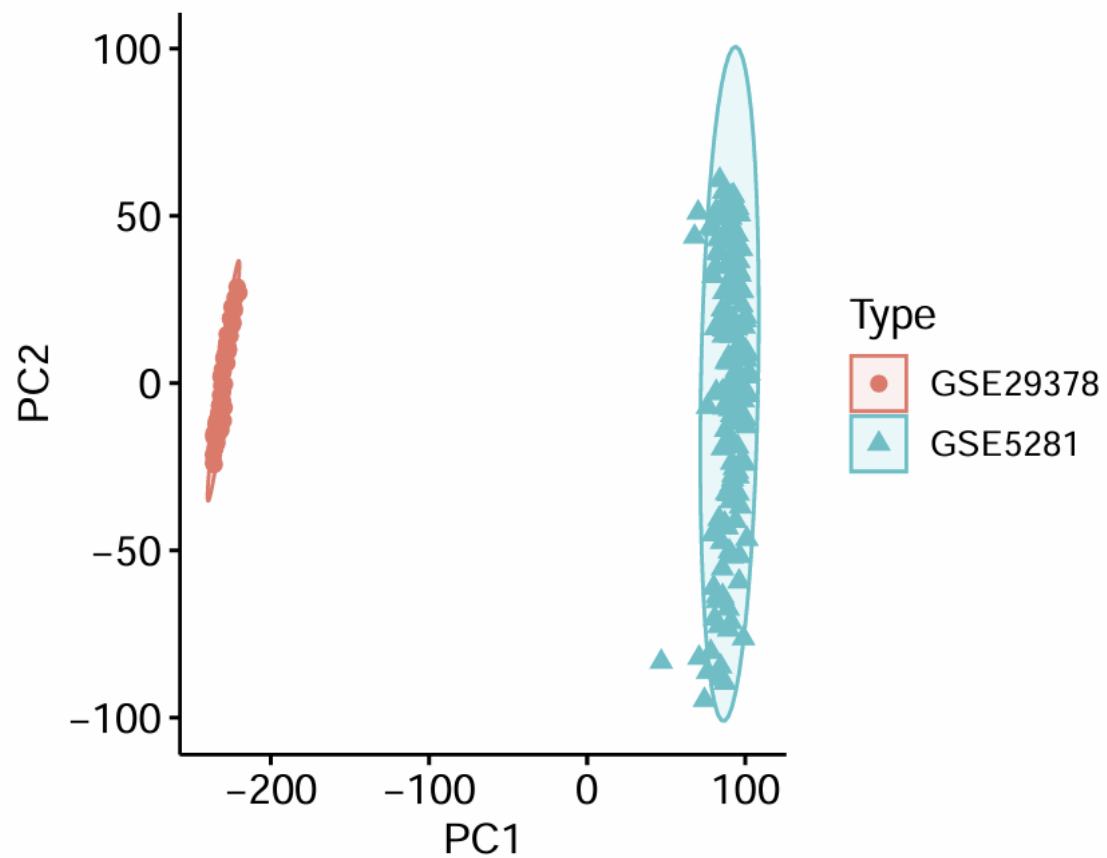
箱线图



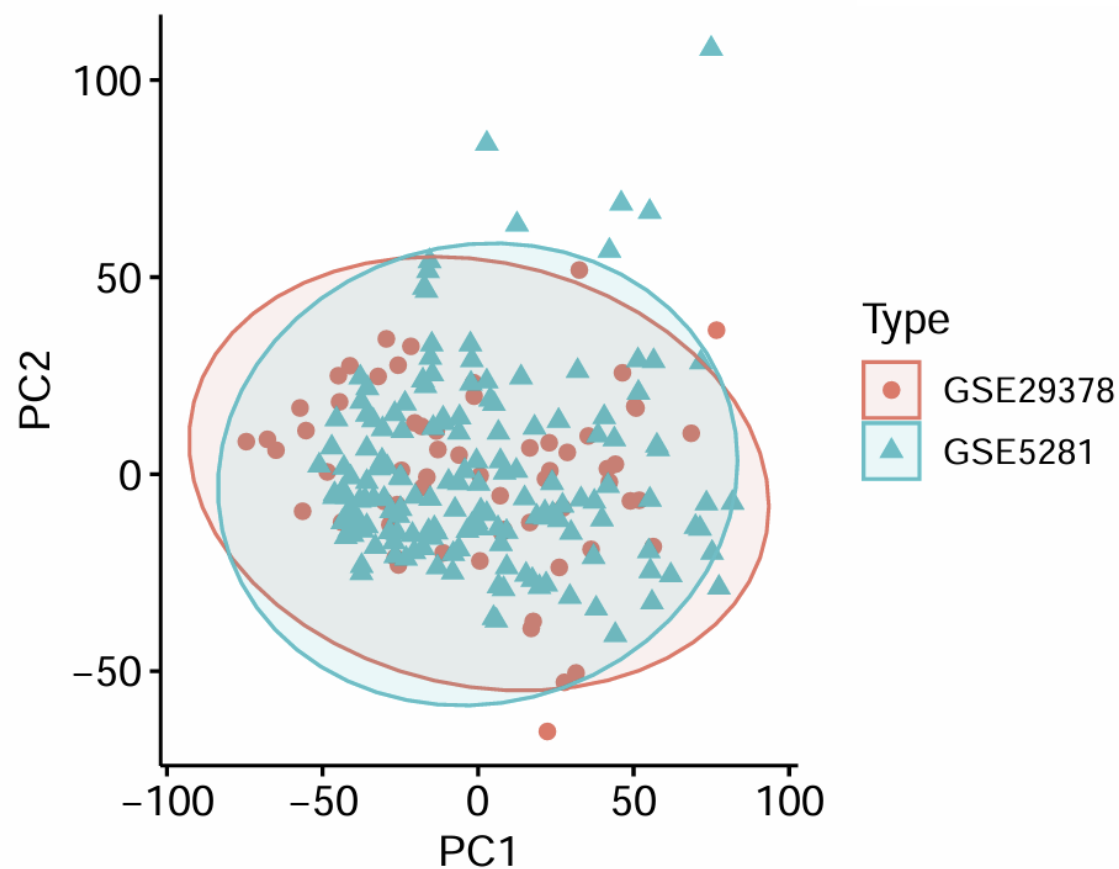
PCA分析



Before batch correction



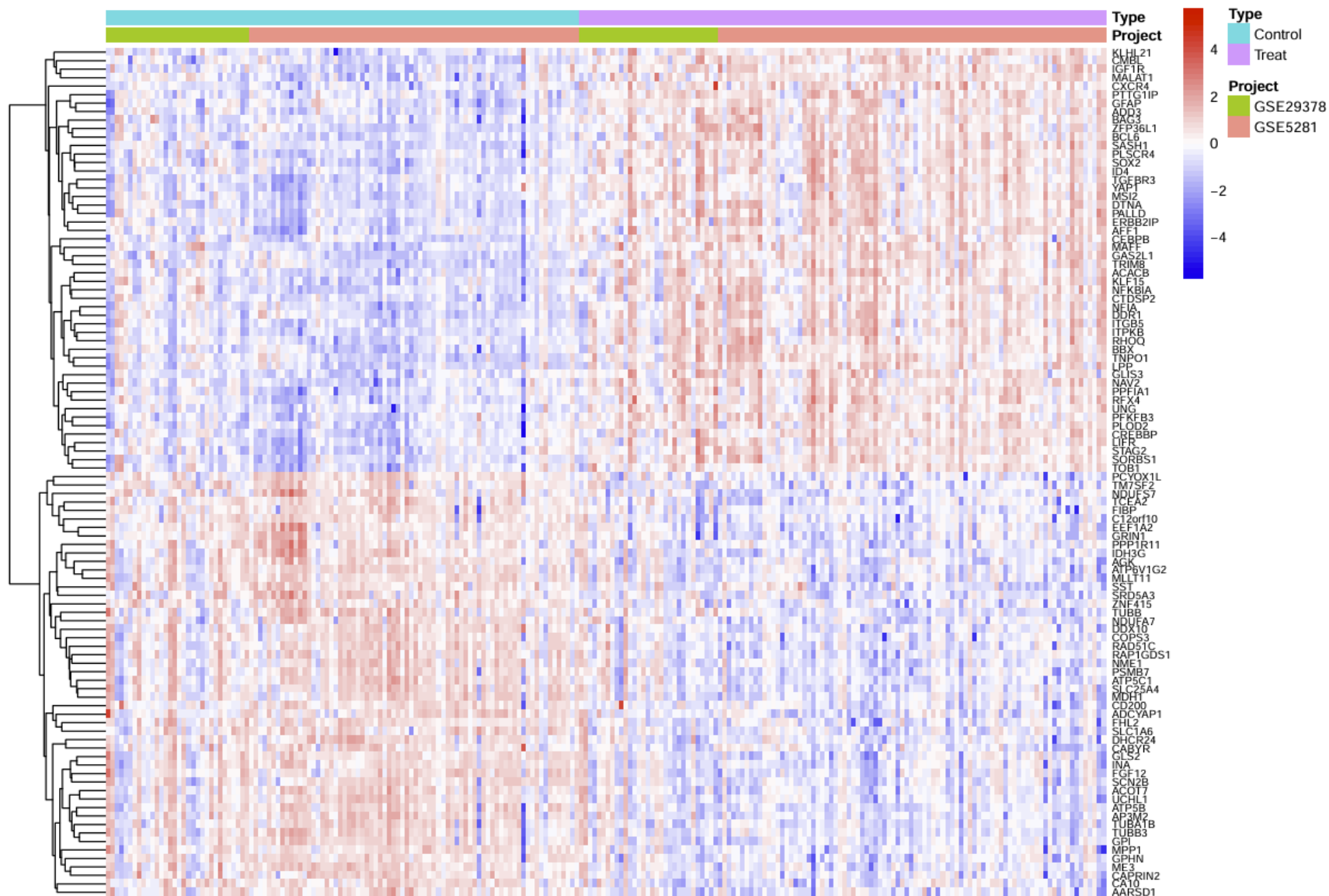
After batch correction



差异分析

id	logFC	AveExpr	t	P.Value	adj.P.Val
GFAP	1.569973	12.18238	10.17667	2.73E-20	2.18E-17
ID3	1.555953	7.642853	7.443911	2.00E-12	1.11E-10
SERPINA3	1.499286	9.219869	6.627174	2.45E-10	6.39E-09
CHST6	1.473444	5.919713	7.600693	7.64E-13	4.80E-11
MAFF	1.452737	8.356968	9.794026	3.99E-19	2.11E-16
DHRS7B	-1.28255	6.500108	-6.47896	5.64E-10	1.33E-08
CHGB	-1.36784	9.63724	-7.1188	1.41E-11	5.81E-10
ARPC1A	-1.40436	8.166968	-6.926	4.38E-11	1.51E-09
C1QTNF4	-1.47726	6.494426	-6.66858	1.93E-10	5.23E-09
SST	-1.86765	7.488238	-7.98347	6.99E-14	5.92E-12

差异基因热图



单细胞数据下载

GEO DataSets

GEO DataSets ▼ (Alzheimer disease single cell) AND "Homo sapiens"[porgn: __txid9606]

Create alert Advanced

Entry type

clear

Summary ▼ 20 per page ▼ Sort by Default order ▼

Send to: ▼

DataSets (0)

✓ Series (132)

Samples (0)

Platforms (0)

Organism

Customize ...

Study type

clear

Expression profiling by array

✓ Expression profiling by high throughput sequencing

Methylation profiling by array

Customize ...

Author

Customize ...

Attribute name

tissue (70)

strain (3)

Customize ...

Search results

Items: 1 to 20 of 132

<< First < Prev Page 1 of 7 Next > Last >>

Filters activated: Series, Expression profiling by high throughput sequencing. [Clear all](#) to show 168 items.

1.

Human stem cell-derived GABAergic interneuron development reveals early emergence of subtype diversity followed by gradual electrochemical maturation [snRNA-seq]

(Submitter supplied) Medial ganglionic eminence-derived inhibitory GABAergic pallial interneurons (MGE-pINs) are essential for cortical development and function. Dysfunction in MGE-pINs is associated with numerous neurological disorders. We developed a human MGE-pIN cell therapy candidate from pluripotent stem cells, which is undergoing clinical trial evaluation for drug-resistant epilepsy (NCT05135091). Here, we performed single nuclei RNA sequencing and analyzed the transcriptomes of xenografted hMGE-pINs over the lifespan of host mice. [more...](#)

Organism: Homo sapiens

Type: Expression profiling by high throughput sequencing

Platform: GPL18573 19 Samples

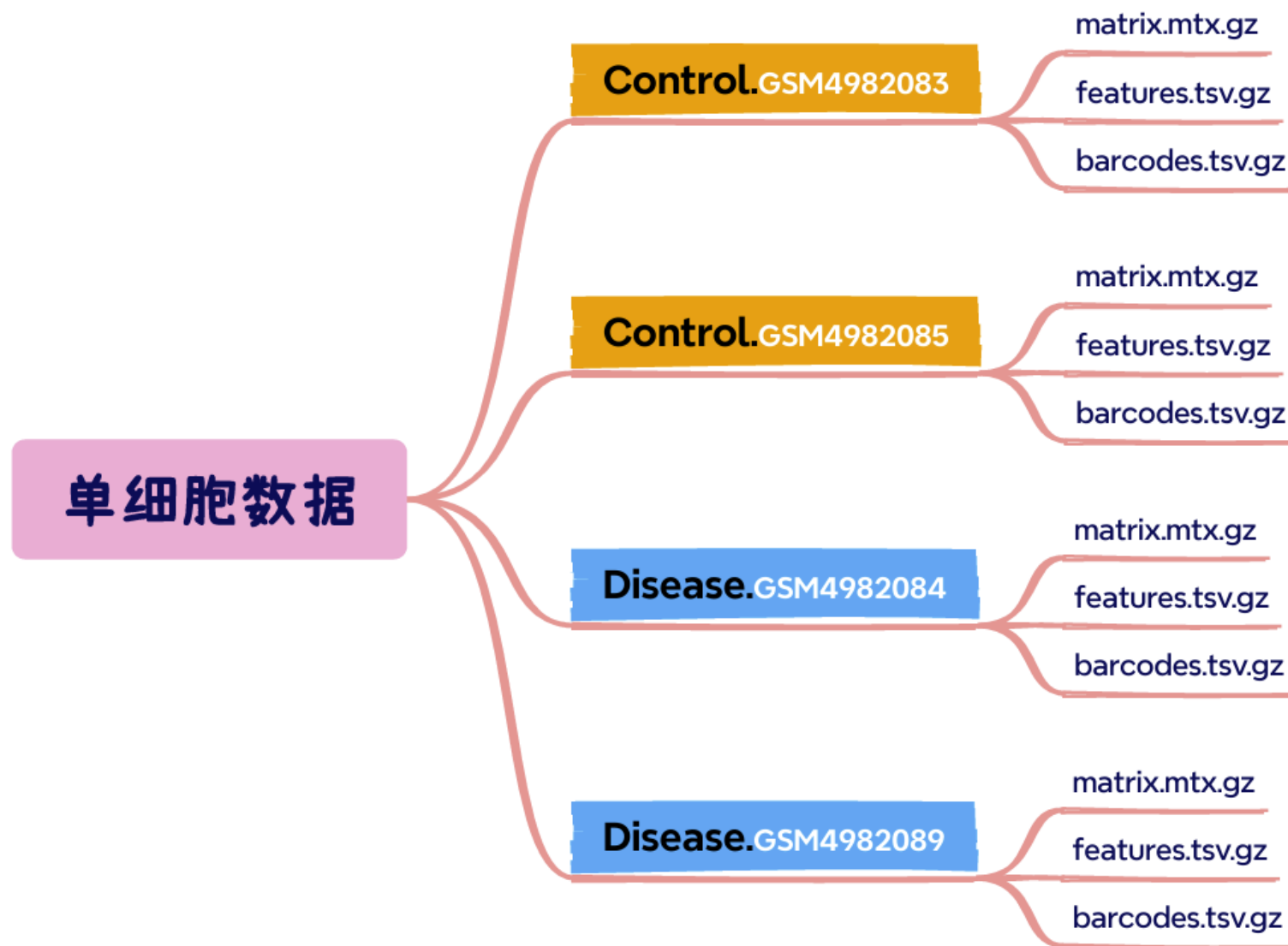
Download data: RDS

Series Accession: GSE283776 ID: 200283776

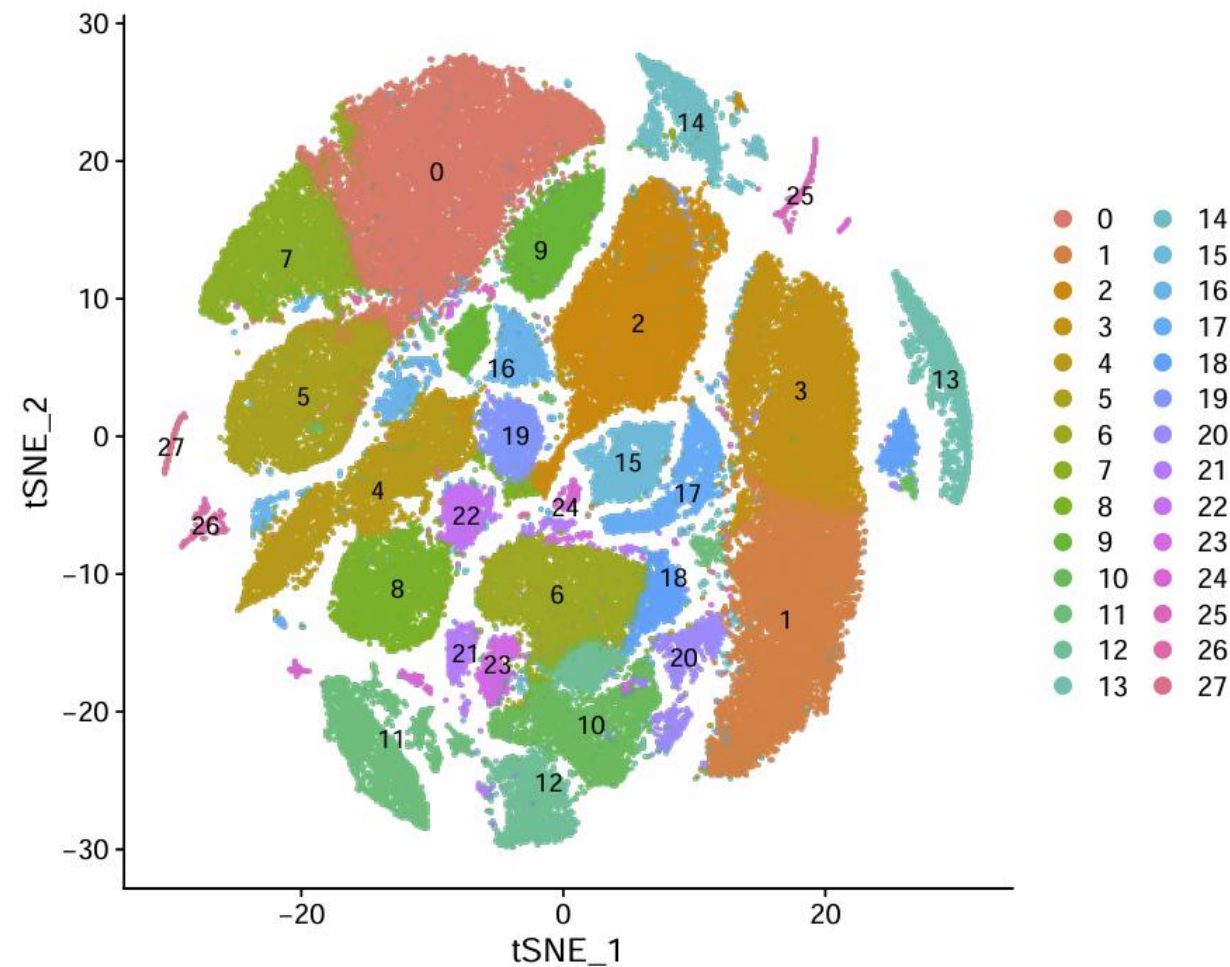
<https://www.ncbi.nlm.nih.gov/geo/>

Alzheimer disease single cell
Alzheimer disease scRNA-seq

单细胞数据整理

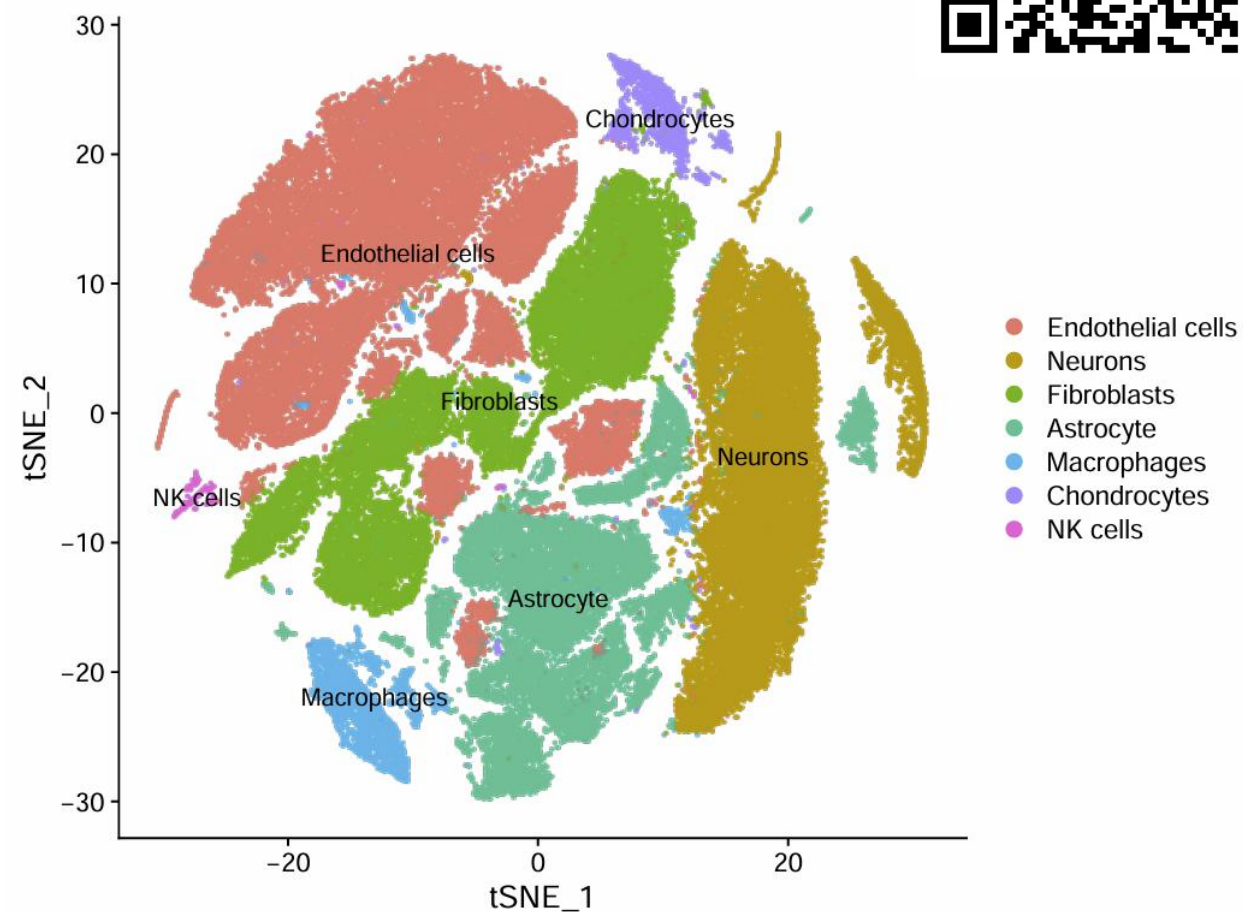
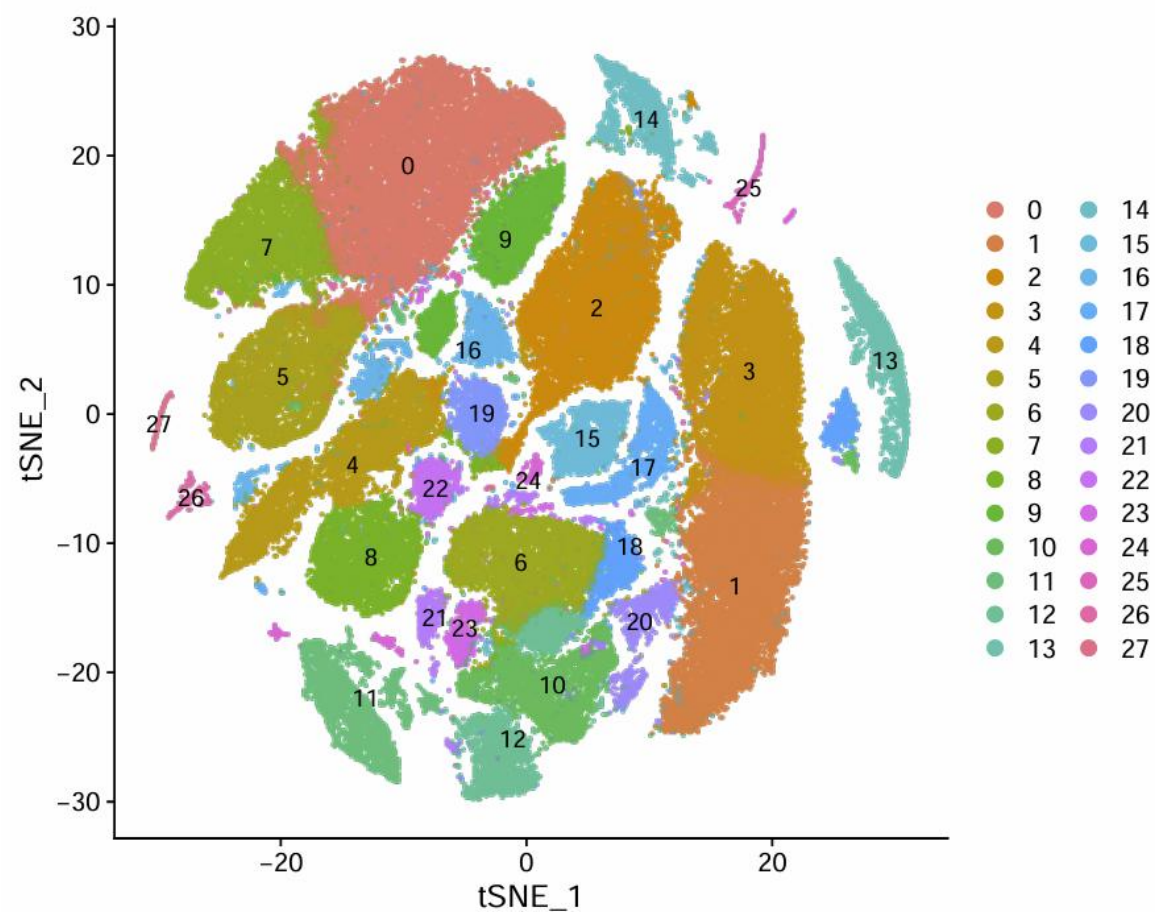


单细胞聚类分析



id	Cluster
Control.GSM4982083_AAACCCAAGAAGTGTT-1	5
Control.GSM4982083_AAACCCATCACTGCTC-1	5
Control.GSM4982083_AAACCCATCCTAACAG-1	0
Control.GSM4982083_AAACCCATCGCAGTCG-1	0
Control.GSM4982083_AAACCCATCTCGGTCT-1	2
Disease.GSM4982084_AAACCCAAGCACCTGC-1	16
Disease.GSM4982084_AAACCCAAGGTTCTAC-1	7
Disease.GSM4982084_AAACCCACAATCTGCA-1	2
Disease.GSM4982084_AAACCCACAGCTACCG-1	4
Disease.GSM4982084_AAACCCACAGGCTTGC-1	12

细胞注释



化合物结构

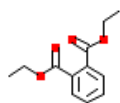
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SEARCH FOR

Diethyl phthalate

Treating this as a text search.

BEST MATCH



[DIETHYL PHTHALATE](#); 84-66-2; Ethyl [phthalate](#); phthalic acid [diethyl](#) ester; Anozol; [diethyl](#) benzene-1,2-dicarboxylate; [Diethyl phthalate](#); Neantine;

Compound CID: 6781

MF: [C₁₂H₁₄O₄](#) MW: 222.24g/mol

IUPAC Name: [diethyl](#) benzene-1,2-dicarboxylate

Isomeric SMILES: CCOC(=O)C1=CC=CC=C1C(=O)OCC

InChIKey: FLKPEMZONWLCSK-UHFFFAOYSA-N

InChI: InChI=1S/C12H14O4/c1-3-15-11(13)9-7-5-6-8-10(9)12(14)16-4-2/h5-8H,3-4H2,1-2H3

Create Date: 2005-03-26

[Summary](#)[Similar Structures Search](#)[Related Records](#)

<https://pubchem.ncbi.nlm.nih.gov/>

Diethyl phthalate

化合物靶点(ChEMBL)



ChEMBL ID	Name	UniProt Accessions	Symbol
CHEMBL1293278	Geminin	O75496	GEMI
CHEMBL1293258	Mothers against decapentaplegic homolog 3	P84022	SMAD3
CHEMBL1871	Androgen Receptor	P10275	ANDR
CHEMBL2034	Glucocorticoid receptor	P04150	GCR
CHEMBL2007625	Isocitrate dehydrogenase [NADP] cytoplasmic	O75874	IDHC
CHEMBL1963	Thyroid stimulating hormone receptor	P16473	TSHR
CHEMBL2362981	TAR DNA-binding protein 43	Q13148	TADBP
CHEMBL1741193	Chromobox protein homolog 1	P83916	CBX1

<https://www.ebi.ac.uk/chembl/>

Diethyl phthalate

化合物靶点(SwissTargetPrediction)

Target	Uniprot ID	Symbol	Probability
Thrombin	P00734	F2	0.060425
Urokinase-type plasminogen activator	P00749	PLAU	0.060425
Carboxylesterase 2	O00748	CES2	0.060425
Cathepsin G	P08311	CTSG	0.060425
Thromboxane-A synthase	P24557	TBXAS1	0.060425
Acetylcholinesterase	P22303	ACHE	0.060425
FK506-binding protein 1A	P62942	FKBP1A	0
Tyrosine-protein kinase JAK3	P52333	JAK3	0
Tyrosine-protein kinase JAK2	O60674	JAK2	0

<http://www.swisstargetprediction.ch/>



Similarity ensemble approach (SEA)

The Similarity ensemble approach relates proteins based on the set-wise chemical similarity among their ligands. It can be used to rapidly search large compound databases and to build cross-target similarity maps.

CCOC(=O)C1=CC=CC=C1C(=O)OCC

Try SEA



Browse Library

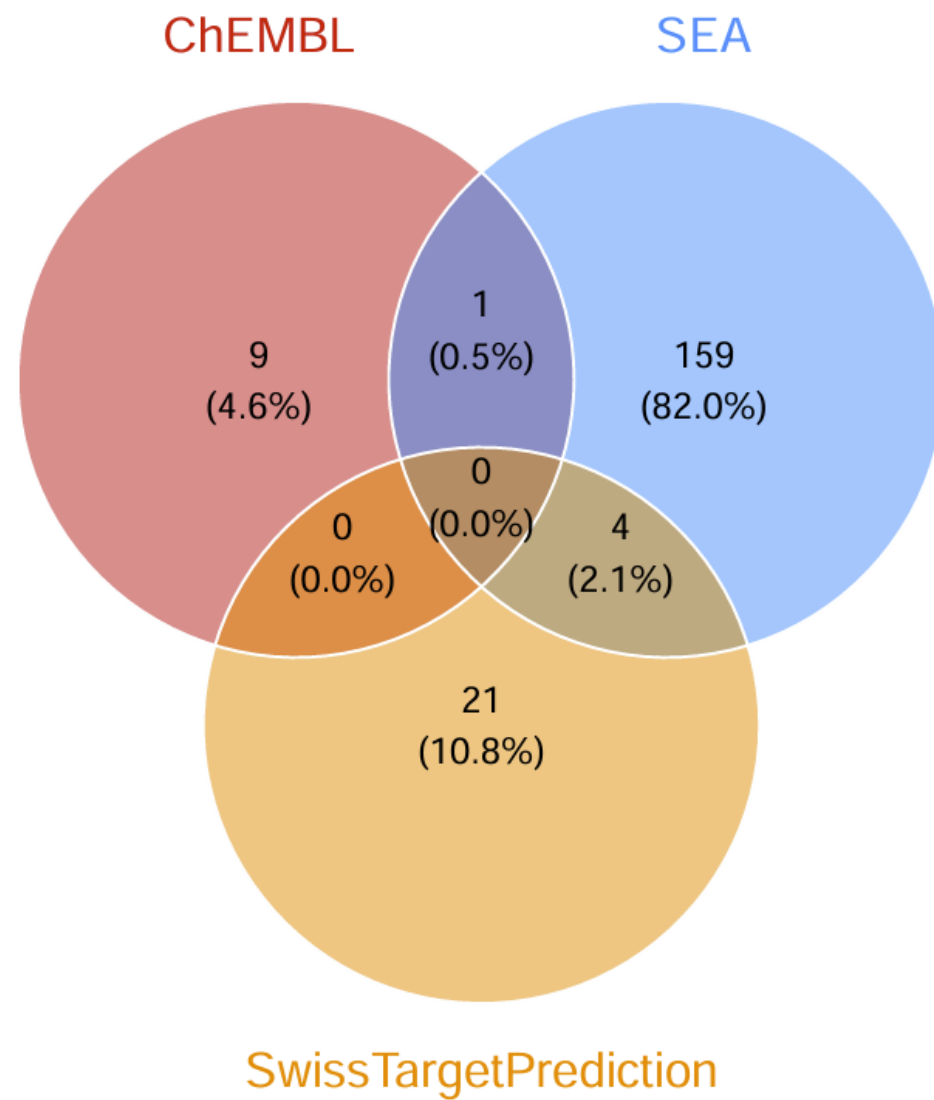
Browse searchable targets from ChEMBL

Search Library

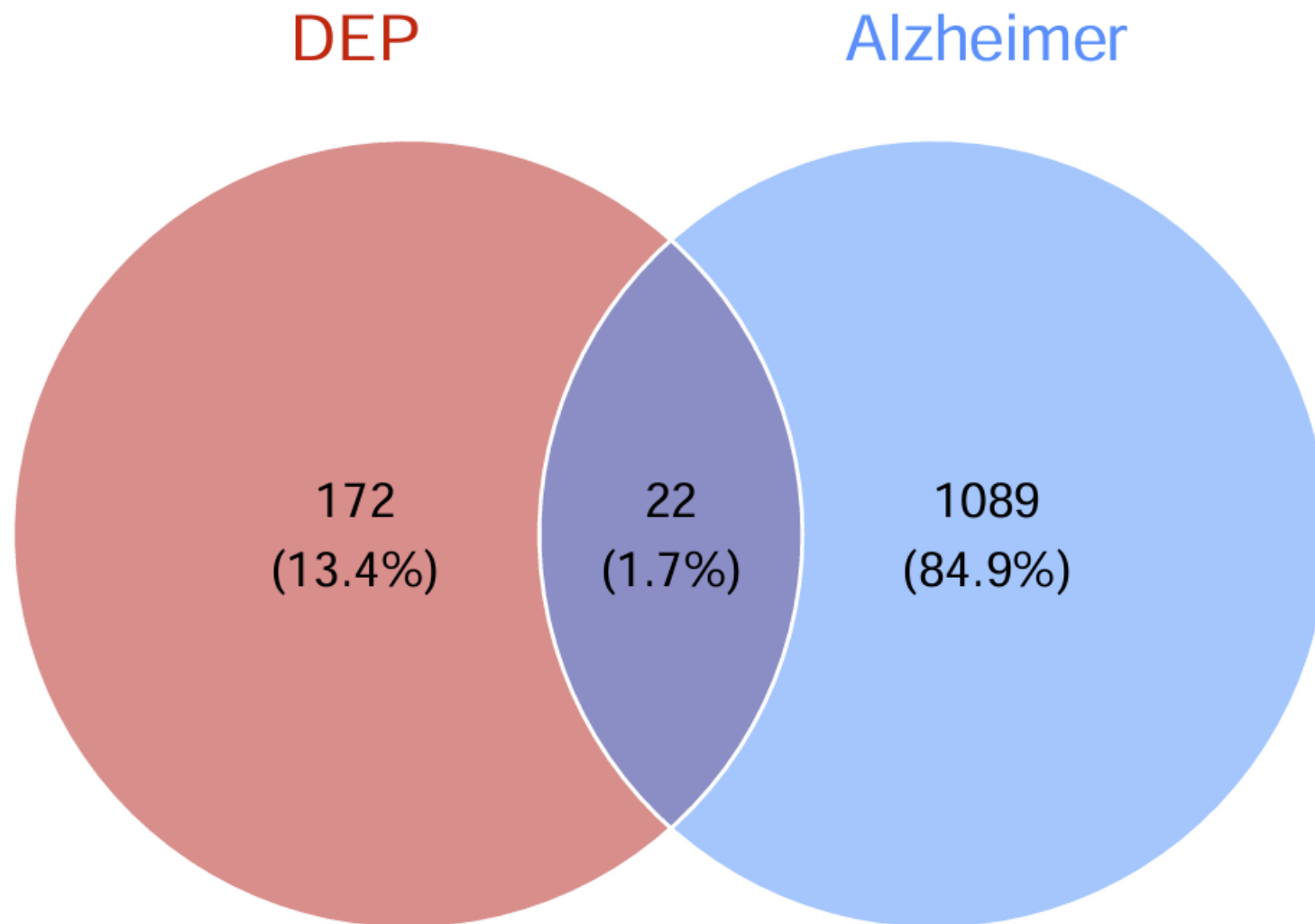
Submit a search of ChEMBL or custom targets

<https://sea.bkslab.org/>

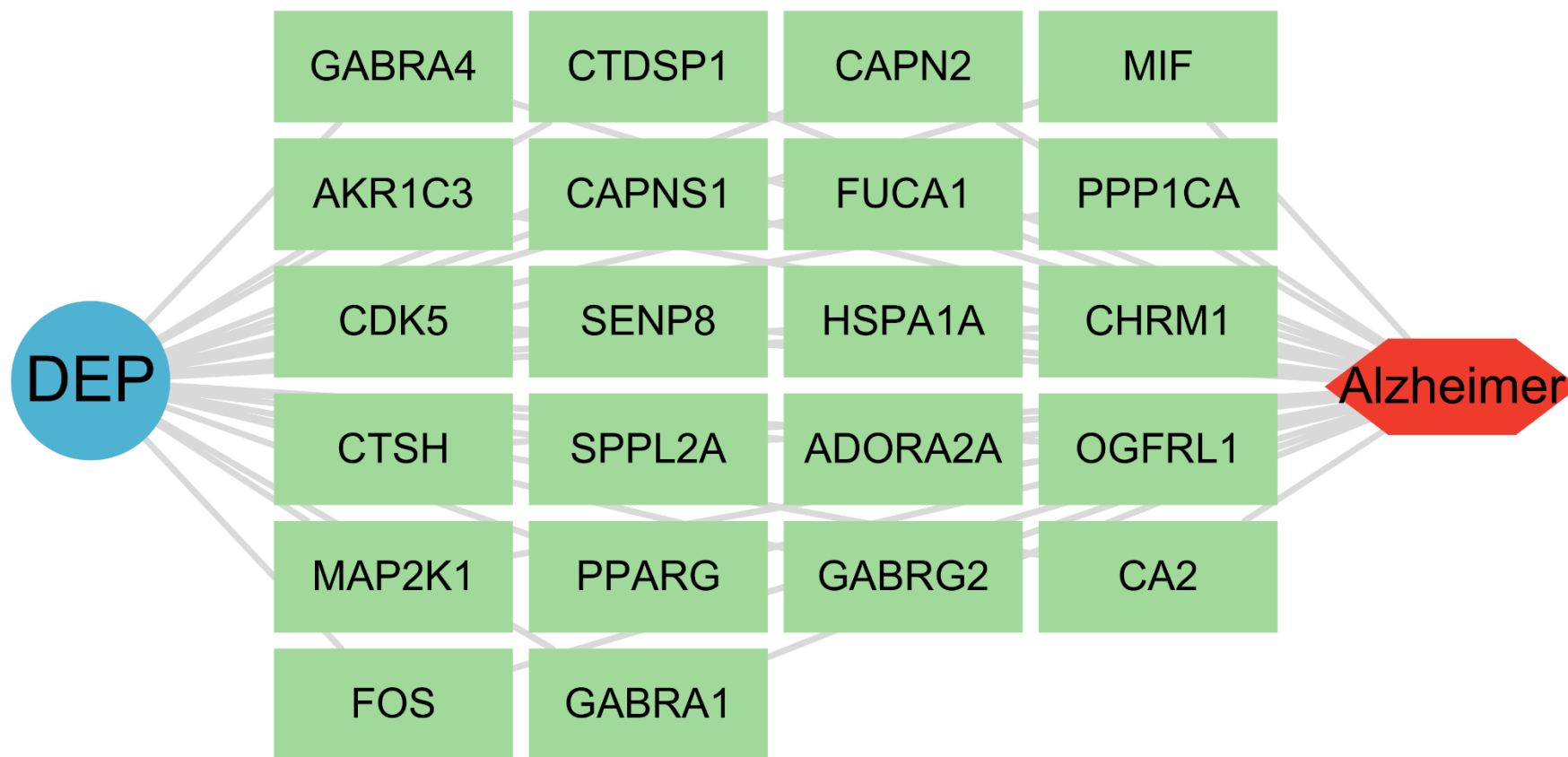
化合物靶基因取并集



化合物靶基因和疾病相关基因取交集



化合物调控网络



<https://cytoscape.org/>

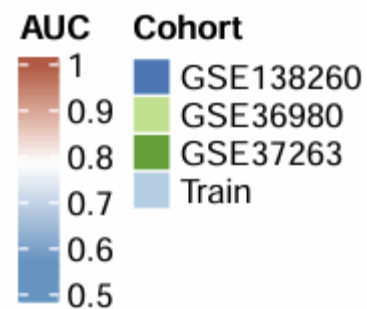
准备机器学习输入文件



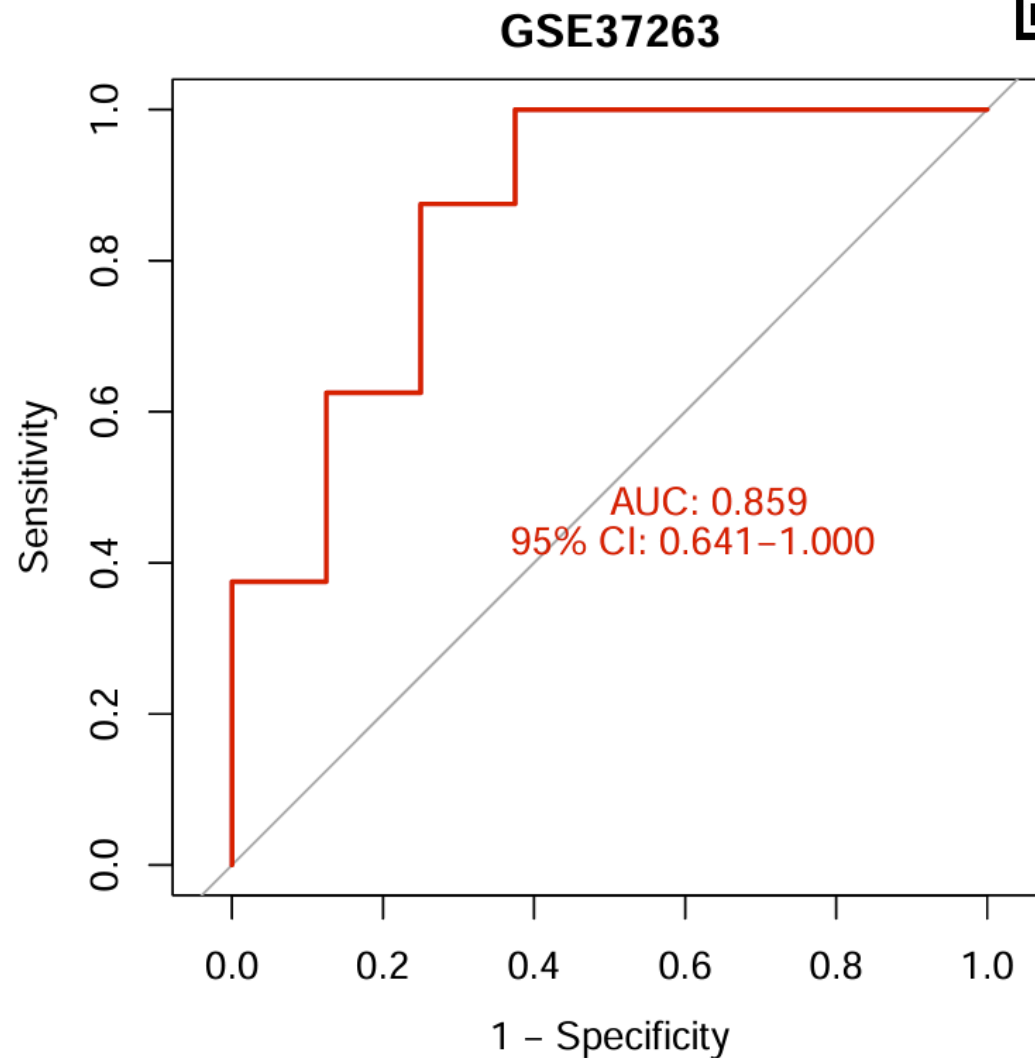
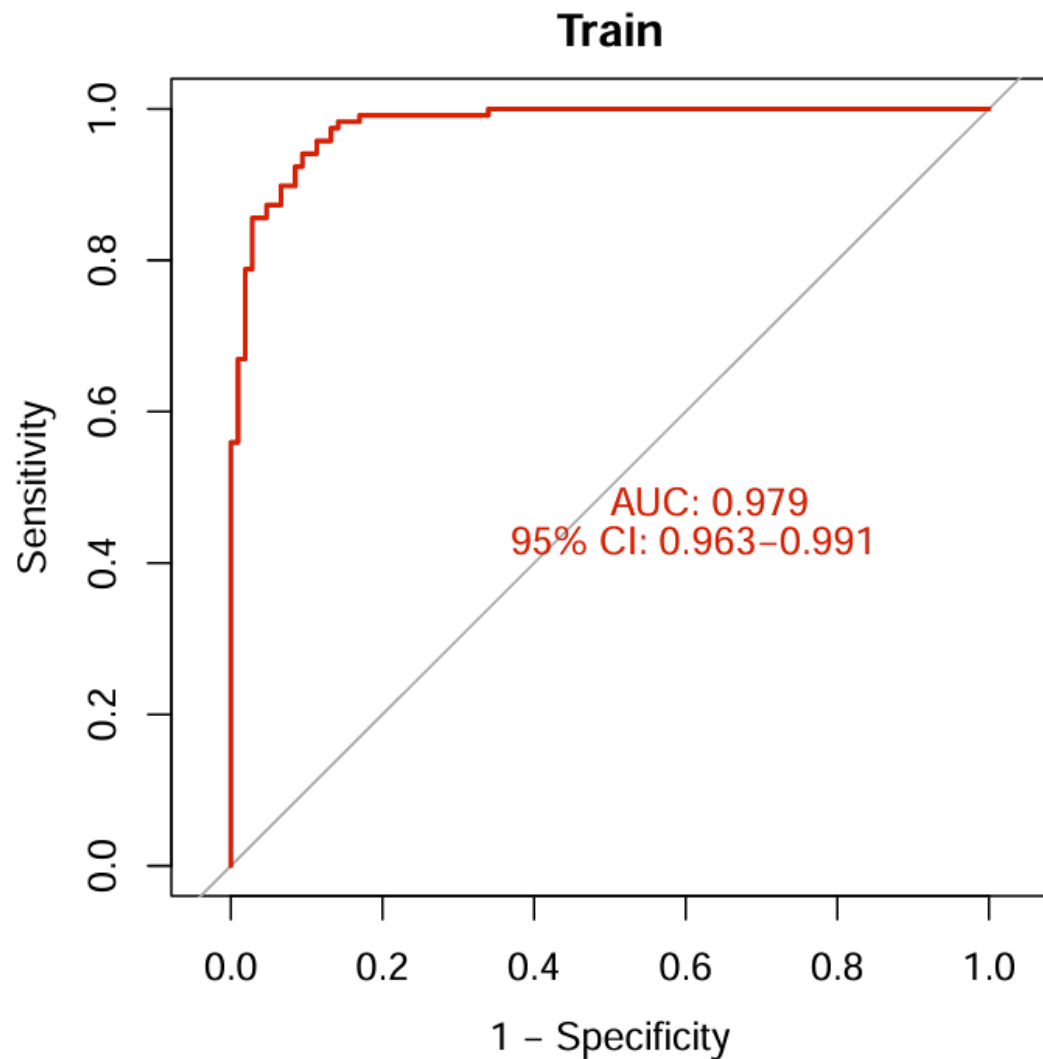
id	ADORA2A	AKR1C3	Type
Train.GSE29378_GSM726117_Control	6.047452	5.967652	0
Train.GSE29378_GSM726088_Control	6.265609	6.097739	0
Train.GSE29378_GSM726100_Control	8.270849	8.845875	0
Train.GSE29378_GSM726140_Control	8.4036	9.132888	0
Train.GSE29378_GSM726109_Control	7.741808	8.25292	0
Train.GSE5281_GSM238763_Treat	8.517051	6.985148	1
Train.GSE5281_GSM238790_Treat	6.162147	7.837848	1
Train.GSE5281_GSM238791_Treat	7.020626	6.572619	1
Train.GSE5281_GSM238792_Treat	7.204904	7.921019	1
Train.GSE5281_GSM238793_Treat	7.836679	7.005896	1

构建机器学习诊断模型

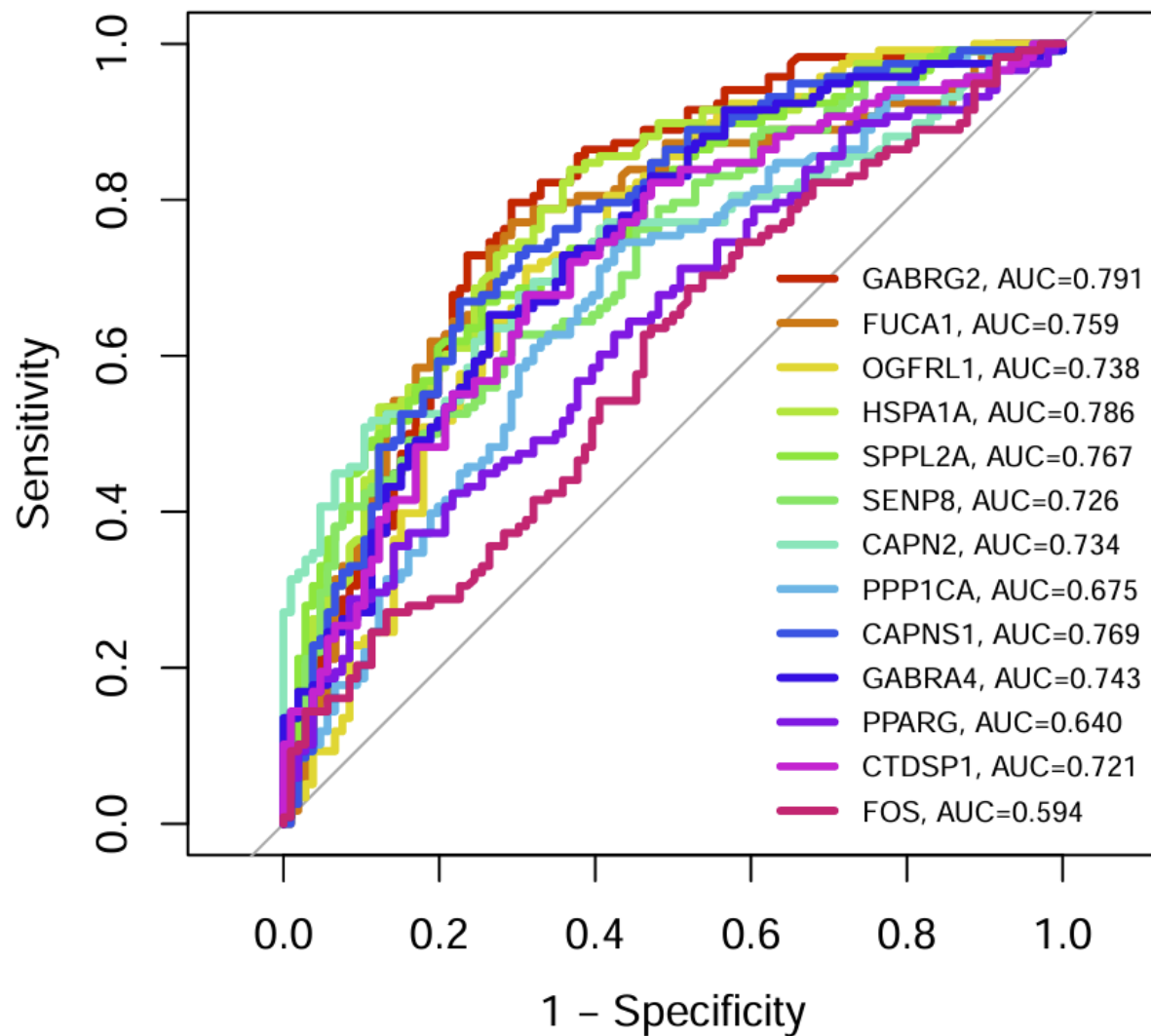
Lasso+GBM	0.979	0.712	0.859	0.859	0.852
Stepglm[backward]+Enet[alpha=0.5]	0.914	0.755	0.869	0.859	0.849
Stepglm[both]+Enet[alpha=0.5]	0.914	0.755	0.868	0.859	0.849
Stepglm[both]+Enet[alpha=0.1]	0.918	0.737	0.866	0.875	0.849
Stepglm[backward]+Enet[alpha=0.1]	0.918	0.737	0.866	0.875	0.849
Stepglm[both]+Enet[alpha=0.4]	0.914	0.752	0.869	0.859	0.849
Stepglm[backward]+Enet[alpha=0.4]	0.913	0.752	0.869	0.859	0.849
Lasso+Stepglm[both]	0.913	0.743	0.863	0.875	0.848
Lasso+Stepglm[backward]	0.913	0.743	0.863	0.875	0.848
glmBoost+Stepglm[both]	0.913	0.743	0.863	0.875	0.848
glmBoost+Stepglm[backward]	0.913	0.743	0.863	0.875	0.848
Lasso+Stepglm[forward]	0.916	0.734	0.868	0.875	0.848
GBM	0.980	0.731	0.853	0.828	0.848
Stepglm[both]+Ridge	0.918	0.734	0.865	0.875	0.848
Stepglm[backward]+Ridge	0.918	0.734	0.865	0.875	0.848
Stepglm[backward]+Enet[alpha=0.8]	0.915	0.749	0.868	0.859	0.848
Stepglm[both]+Enet[alpha=0.7]	0.915	0.749	0.868	0.859	0.848
Stepglm[backward]+Enet[alpha=0.6]	0.914	0.749	0.868	0.859	0.848
Stepglm[backward]+GBM	0.986	0.724	0.836	0.844	0.848
Stepglm[both]+GBM	0.984	0.721	0.841	0.844	0.847
Stepglm[backward]+Enet[alpha=0.7]	0.915	0.746	0.867	0.859	0.847
Stepglm[both]+Lasso	0.914	0.746	0.867	0.859	0.847
Stepglm[backward]+Lasso	0.914	0.746	0.867	0.859	0.847
Stepglm[backward]+Enet[alpha=0.9]	0.914	0.746	0.866	0.859	0.846
Stepglm[both]+glmBoost	0.915	0.743	0.865	0.859	0.846
Stepglm[backward]+glmBoost	0.915	0.743	0.865	0.859	0.846
glmBoost+Lasso	0.914	0.724	0.868	0.875	0.846
Stepglm[backward]+Enet[alpha=0.2]	0.913	0.759	0.867	0.844	0.845
glmBoost+Enet[alpha=0.9]	0.914	0.724	0.867	0.875	0.845
glmBoost+Stepglm[forward]	0.914	0.724	0.867	0.875	0.845
Enet[alpha=0.7]	0.914	0.724	0.860	0.875	0.843



绘制模型的ROC曲线



模型基因的ROC曲线



列线图



Points

GABRG2

FUCA1

OGFRL1

HSPA1A

SPPL2A

SENP8

CAPN2

PPP1CA

CAPNS1

GABRA4

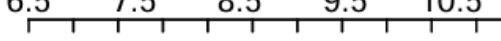
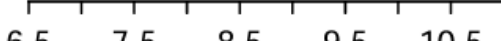
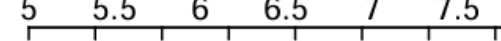
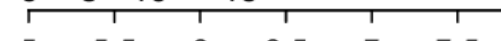
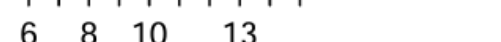
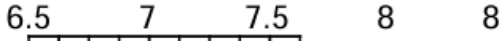
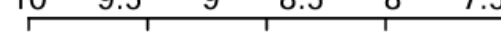
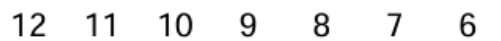
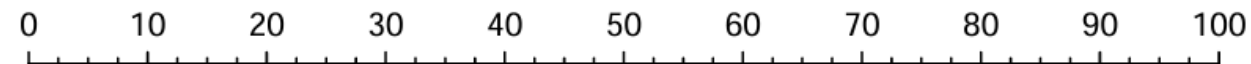
PPARG

CTDSP1

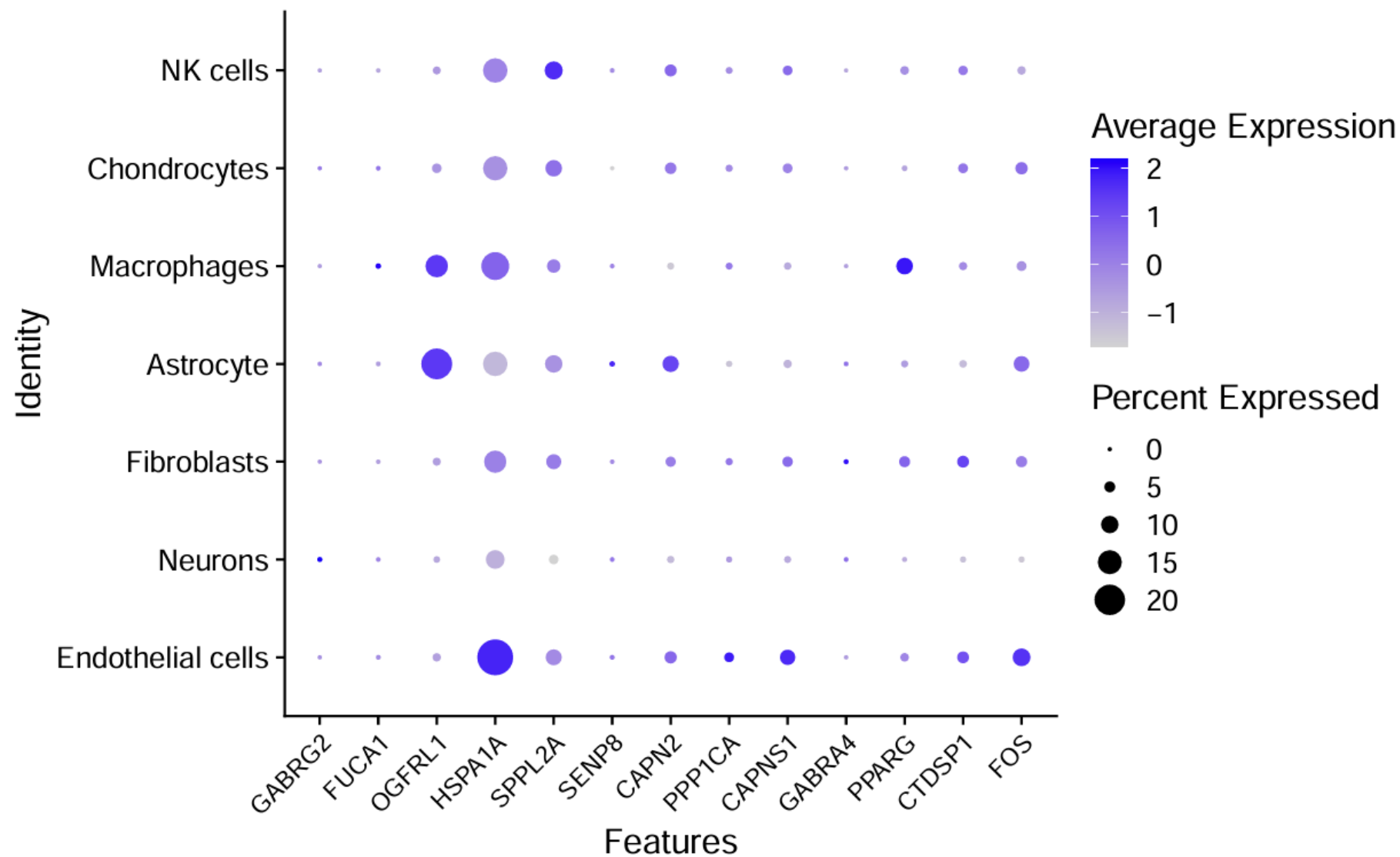
FOS

Total Points

Risk of Disease



模型基因气泡图

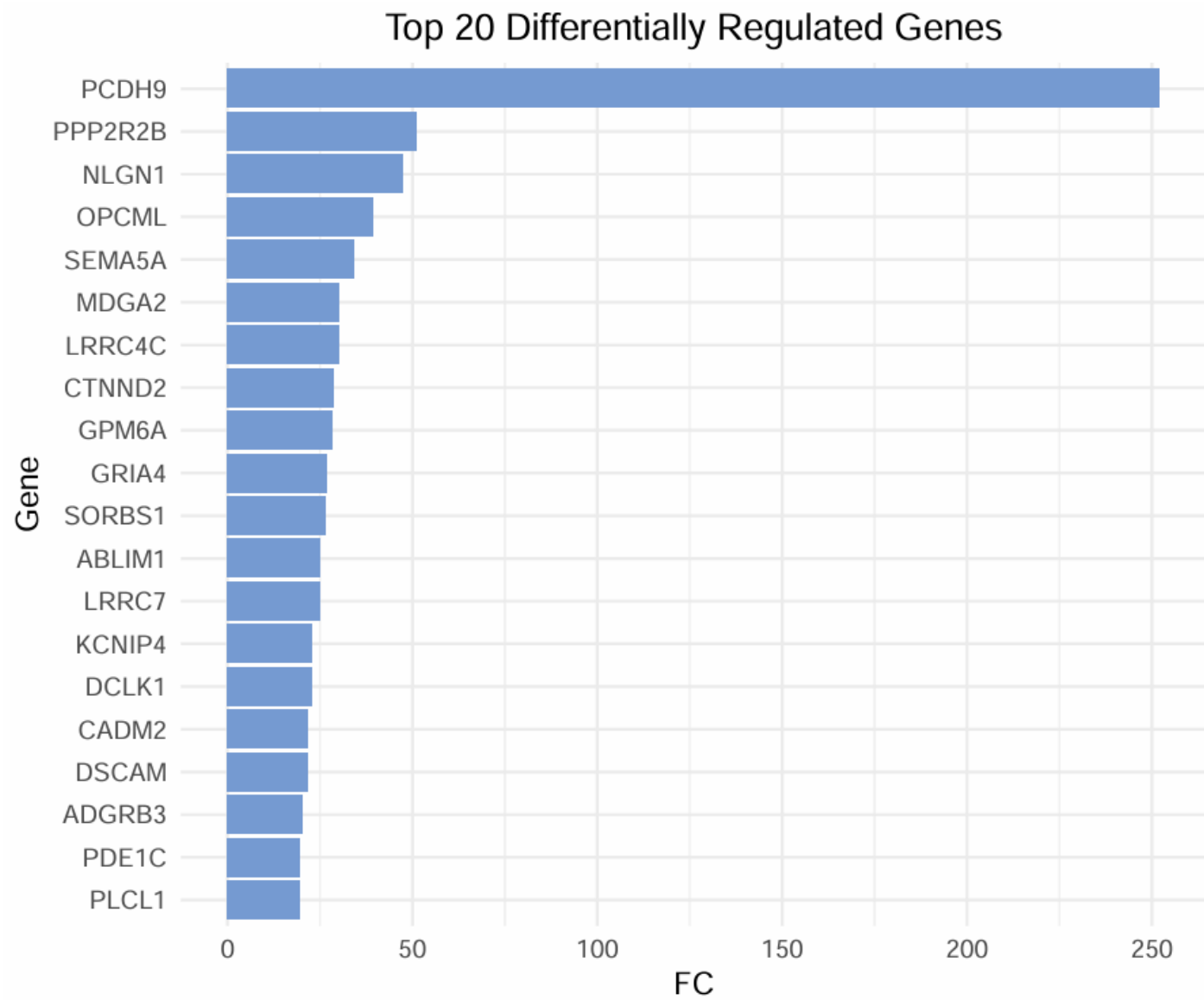


模拟基因敲除(scTenifoldKnk)



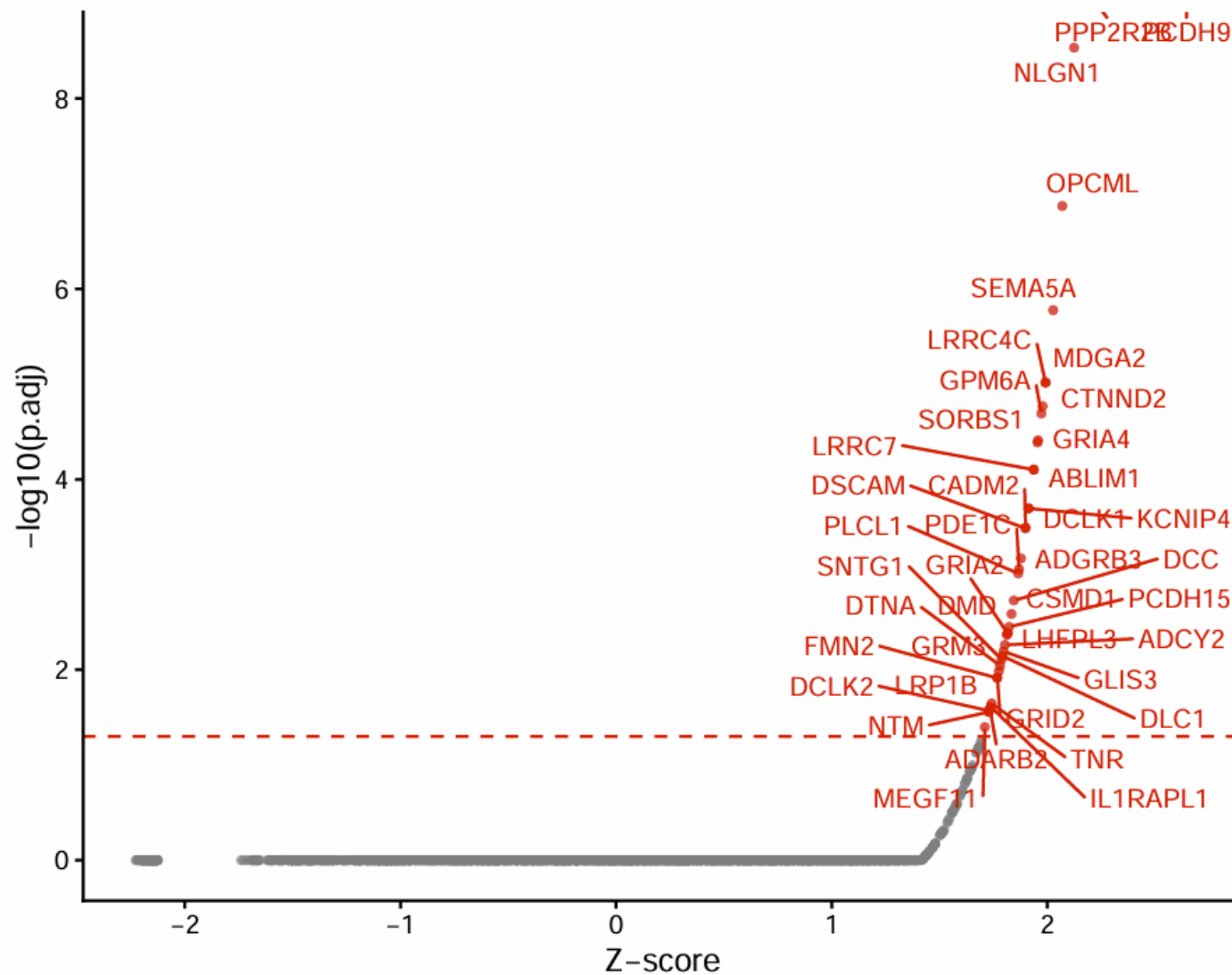
gene	distance	Z	FC	p.value	p.adj
PCDH9	4.14E-06	2.655235	252.1507	8.82E-57	8.97E-54
PPP2R2B	1.86E-06	2.146778	51.17931	8.43E-13	5.71E-10
NLGN1	1.79E-06	2.123734	47.41135	5.75E-12	2.92E-09
OPCML	1.64E-06	2.069059	39.48122	3.31E-10	1.35E-07
SEMA5A	1.52E-06	2.02676	34.215	4.93E-09	1.67E-06
MDGA2	1.44E-06	1.992026	30.38866	3.54E-08	9.61E-06
LRRC4C	1.43E-06	1.990779	30.2591	3.78E-08	9.61E-06
CTNND2	1.40E-06	1.977603	28.92038	7.54E-08	1.70E-05
GPM6A	1.39E-06	1.97206	28.37393	1.00E-07	2.03E-05
GRIA4	1.35E-06	1.956993	26.93674	2.10E-07	3.89E-05

柱状图



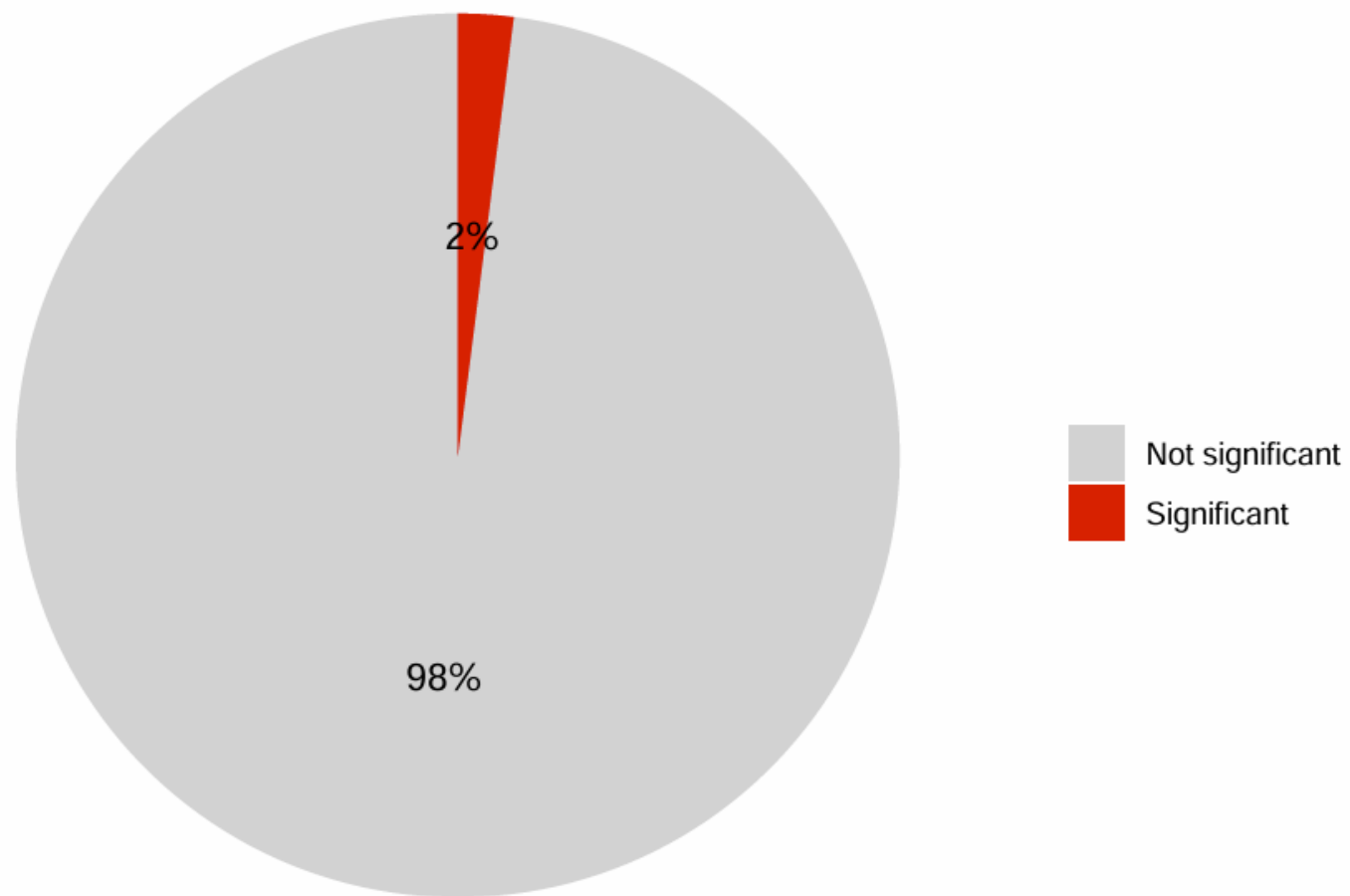


火山图

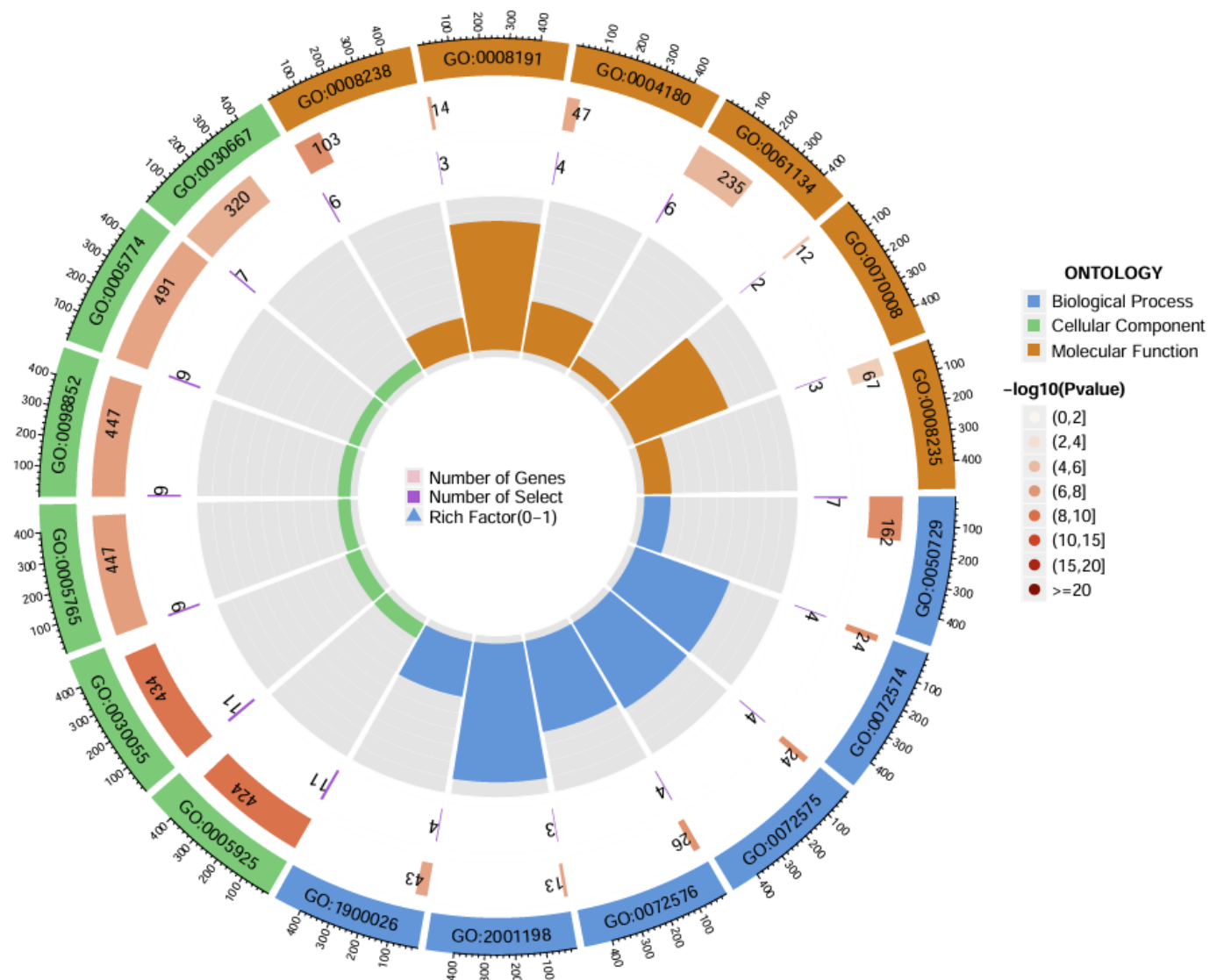


饼图

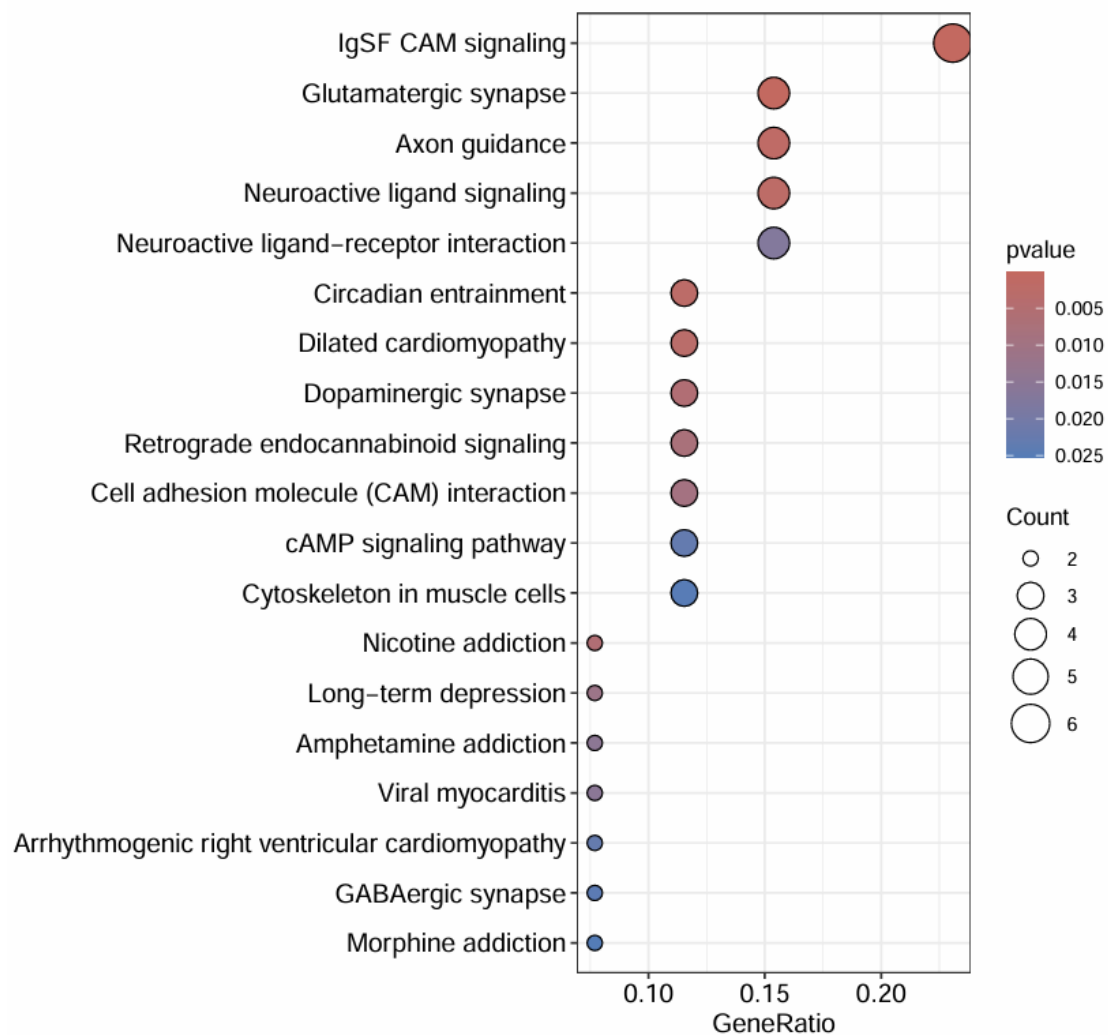
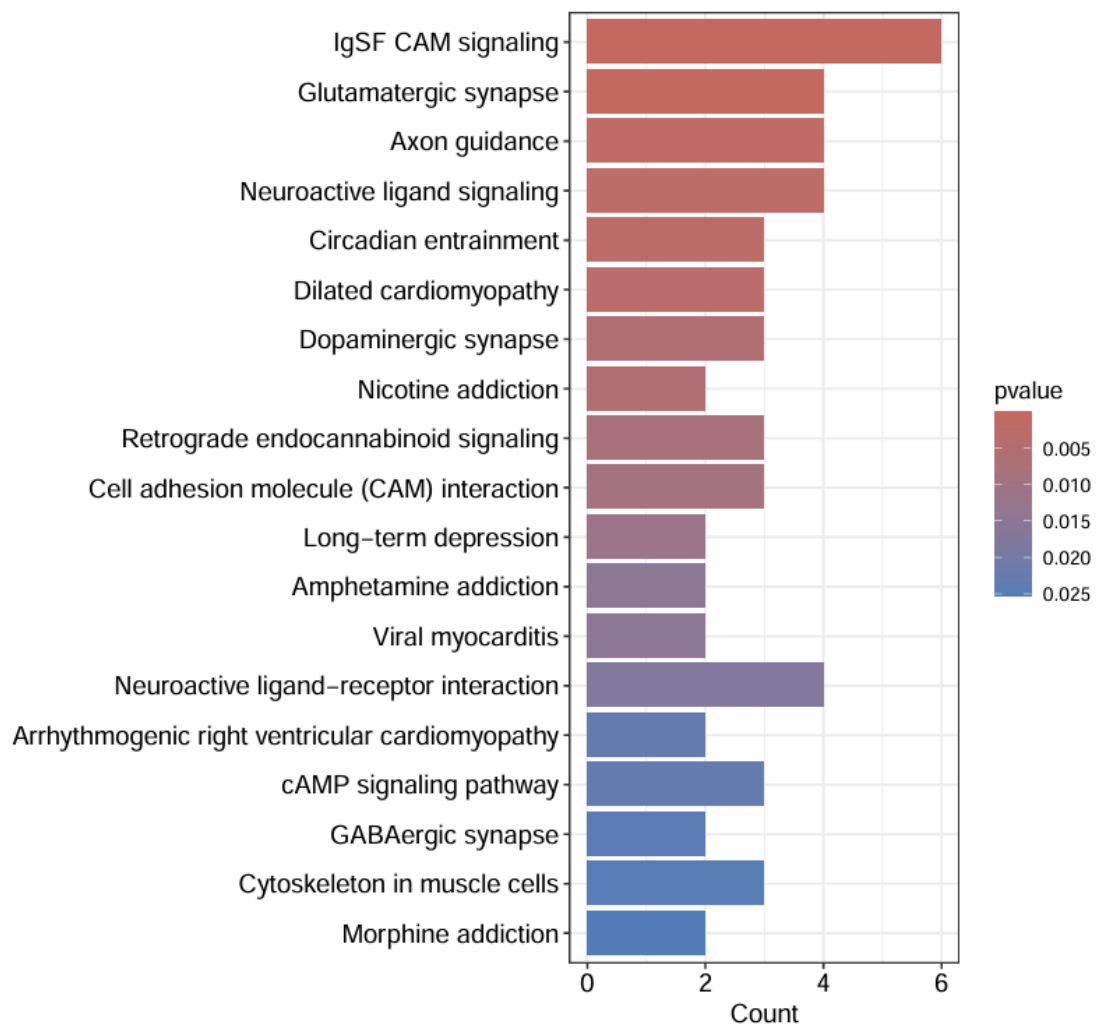
Proportion of Significant Genes



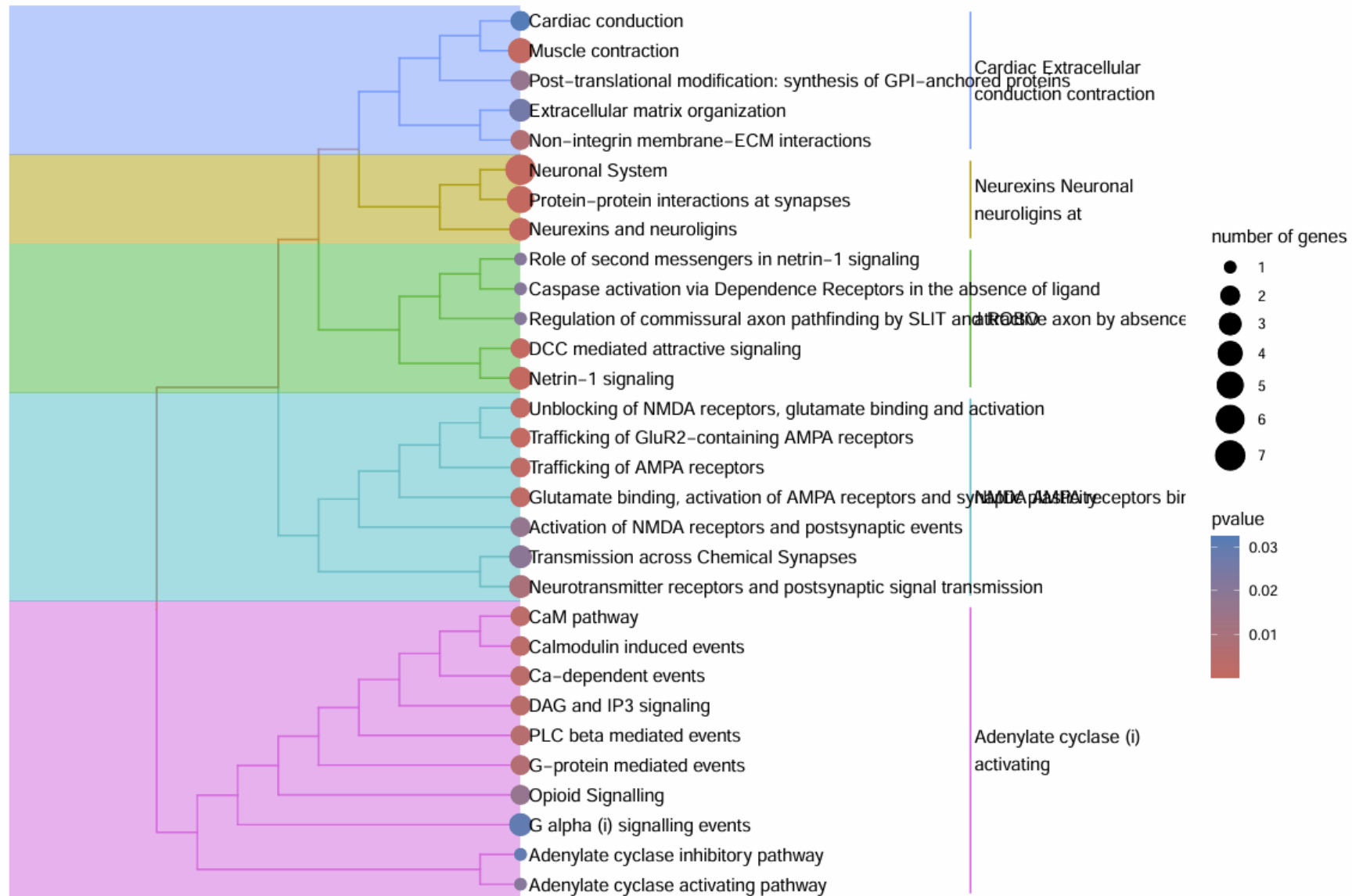
GO富集分析(功能)



KEGG富集分析(通路)



Reactome富集分析



分子对接数据(蛋白的结构)

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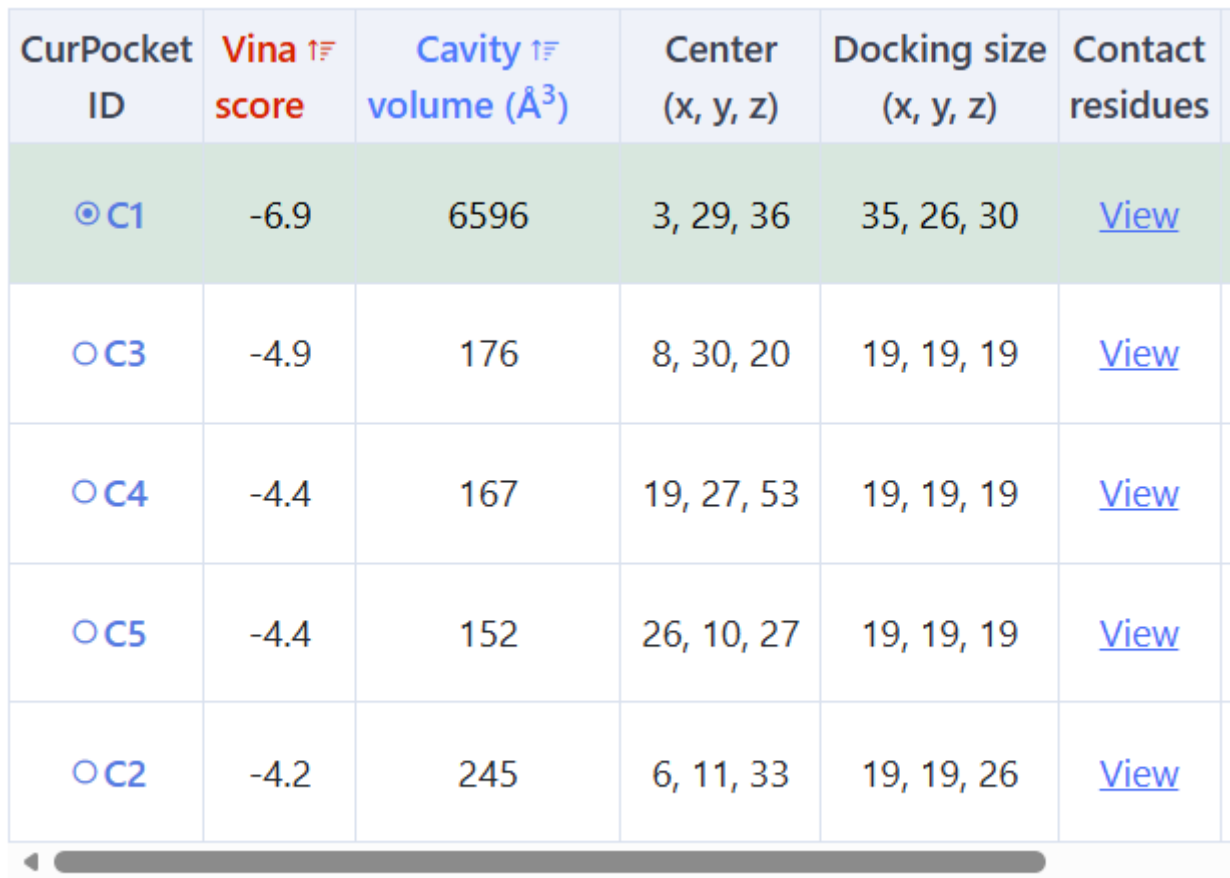
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Thanks!!!



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